

Alpha-One Intermittency: A Common Complex Domain and Induced Fredholm Product

HCS-C381 theorem and reproducibility package

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Abstract

We carry an exact first-return construction for alpha-one intermittency into one fixed complex disk. All countably many inverse return branches map that disk into a common strictly interior disk, and a reciprocal-growth identity gives uniformly summable quadratic derivative decay. A direct Hardy-basis rank-one expansion proves trace-class convergence for every original-time parameter in the closed unit disk. The induced Fredholm determinant is entire in the return-count parameter and has a derivative-weighted primitive product with both clocks recorded explicitly. Its trace denominator accounts for all repetitions, while the neutral fixed point remains outside the induced hyperbolic ledger. The defining branch series fails beyond the unit disk, which is not a claim of a natural boundary. The construction is an explicit source-system proof, with classical intermittency and inducing credited to their literature.

中文摘要

本文把一阶间歇映射的精确首次返回提升到一个固定复圆盘。
全部可数逆返回分支均把该圆盘送入同一严格内含圆盘；倒数增长与导数望远镜乘积给出统一的二次衰减。
在固定哈代空间的单项式基中直接构造秩一展开，从而证明原始时间参数闭单位圆盘内的迹类收敛。
诱导弗雷德霍姆行列式对返回次数参数为整函数，并在初始收敛域具有精确的本原轨道乘积。
乘积同时记录返回次数与原始迭代次数，稳定性分母保留全部重复轨道的贡献。
中性固定点从诱导扩张轨道中单独分离，不能把其单位乘子塞入通常的平迹分母。
定义所用分支级数在单位圆盘外失败，但这不被误称为禁止一切解析延拓的自然边界。
本文给出可审查的源系统证明，并保留经典间歇性与诱导方法的文献归属。

Keywords: inducing; complex domain; Hardy space; trace class; Fredholm determinant; primitive orbit product
关键词：诱导；复域；哈代空间；迹类；弗雷德霍姆行列式；本原轨道乘积

1 A fixed endpoint and the complete periodic census

Fix

$$T(x) = \begin{cases} x + 2x^2, & 0 \leq x \leq 1/2, \\ 2x - 1, & 1/2 < x \leq 1. \end{cases} \quad g(z) = \frac{\sqrt{1+8z}-1}{4}, \quad h(z) = \frac{1+z}{2}. \quad (1)$$

The endpoint $1/2$ belongs to the left branch, so $T(1/2) = 1$. Its assignment changes a countable boundary orbit set relative to the other common convention and will matter for literal return intervals. The complex square root is chosen with positive real part on $\text{Re } z > 0$. The neutral point is zero. Original time counts one application of T .

Theorem 1 (All periodic words, with the neutral orbit separated). *For every integer $n \geq 1$, T^n has exactly 2^n fixed points. Binary length- n words give a bijection with these based points. The number of primitive cycles of period n is*

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}. \quad (2)$$

Here μ is the Möbius function. Only the all-zero orbit is neutral. Every other periodic orbit has positive multiplier at least two. Least word period and least orbit period coincide.

Proof. On $[0, 1]$, the maps g, h are increasing, $0 < g' \leq 1$, and $h' = 1/2$. Any inverse composition containing h is a contraction with Lipschitz constant at most $1/2$, so it has exactly one fixed point. The all- g composition has only zero because $g(x) < x$ for positive x ; the all- h composition has fixed point one.

Before reading forward itineraries, exclude boundary ambiguity. If an intermediate point of a mixed inverse cycle were zero, the inverse step producing it would be g with zero input. Iterating this implication around

the cycle forces every branch to be g . If it were one, the step would be h with input one, forcing the all- h cycle. Thus mixed cycles avoid zero and one. They also avoid $1/2$, which can arise only as $g(1)$ or $h(0)$. Their forward itineraries are therefore unambiguous and agree with the prescribed branch composition. Two different based words cannot produce the same point because the deterministic forward orbit has only one itinerary. Conversely every fixed point solves the equation for its inverse word. This proves the complete count and identifies least periods.

Möbius inversion of the primitive-power decomposition gives (2). Every nonzero periodic word visits the right branch, where $T' = 2$, while each left derivative is $1 + 4x \geq 1$. Multiplying these positive derivatives gives the multiplier statement. Under a repetitions, the period is multiplied by a and the multiplier becomes Λ^a . $\square$

Each branch preserves orientation. No time-reversal symmetry is declared for this noninvertible interval map.

2 Every first-return branch and the sharp tail

Set $Y = (1/2, 1]$, let τ be the first positive return time to Y , and write $R = T^\tau$. Define

$$h_n(z) = \frac{1 + g^{n-1}(z)}{2} \quad (n \geq 1), \quad a_m = g^m(1/2) \quad (m \geq 0). \quad (3)$$

Theorem 2 (Complete return partition and reference-measure tail). *Every point of Y has a finite first return, and*

$$\{y \in Y : \tau(y) = n\} = h_n(Y), \quad R \circ h_n = \text{id}_Y. \quad (4)$$

These left-open, right-closed intervals partition Y . For $n \geq 1$,

$$\text{Leb}\{y \in Y : \tau(y) > n\} = \frac{a_{n-1}}{2} \sim \frac{1}{4n}. \quad (5)$$

Under normalized Lebesgue measure on Y , the tail is asymptotic to $1/(2n)$ and its mean is infinite.

Proof. After a point of Y takes the right branch it is positive. While it remains in $(0, 1/2]$, the left branch increases it strictly. If it never escaped, it would converge to a positive fixed point of $x + 2x^2$, which is impossible. The inverse of a return of length n first reverses $n - 1$ left steps and then one right step, giving (3). The preceding inverse left iterates lie in $(0, 1/2]$, which proves the first-return assertion. Moreover $g(1) = 1/2$ gives $h_{n+1}(1) = h_n(1/2)$. The left-branch assignment at $1/2$ makes the shared upper endpoints return on the claimed iterate. These adjacent intervals exhaust Y as their left endpoints tend to $1/2$.

Their endpoints give the exact tail in (5); for $n = 1$ it is $(1/2, 3/4]$, of length $1/4$. The recurrence $a_{m-1} = a_m + 2a_m^2$ gives

$$\frac{1}{a_m} - \frac{1}{a_{m-1}} = \frac{2}{1 + 2a_m}, \quad \frac{1}{2m + 2} \leq a_m \leq \frac{1}{m + 2}. \quad (6)$$

The increment lies between one and two. Using the upper bound in $2/(1 + 2a_m) = 2 - 4a_m/(1 + 2a_m)$ and summing yields $a_m^{-1} = 2m + O(\log(m + 2))$. Therefore $a_m = (2m)^{-1} + O(\log m/m^2)$ and (5) follows. Normalization doubles the tail, and the harmonic-series comparison gives the infinite mean. This is a reference-Lebesgue statement; no stationary induced density has been assumed. $\square$

Corollary 1 (Primitive induction preserves the original clock). *Nonneutral primitive T cycles correspond bijectively to primitive R cycles. If an induced cycle has return period r_p , return times $n_1, \dots, n_{r_p}$, and total multiplier Λ_p , its original period is $N_p = \sum_j n_j$ and its multiplier remains Λ_p .*

Proof. Every nonneutral periodic word contains a right symbol and hence visits Y . Successive gaps between such visits determine the induced itinerary. Conversely (3) reconstructs the original cycle. Changing the chosen visit rotates the induced word. Thus no additional multiplicity equal to the number of returns is introduced. A smaller original or induced period would force a smaller period on the other side. $\square$

The unweighted zeta and its neutral-point removal are consequently

$$Z_T(z) = \frac{1}{1 - 2z}, \quad Z_{T, \text{nonneutral}}(z) = \frac{1 - z}{1 - 2z}, \quad |z| < 1/2. \quad (7)$$

Indeed the first is the exponential series of the exact fixed counts; the second removes the single primitive fixed point. The elementary renewal identity $1 - \sum_{n \geq 1} z^n = (1 - 2z)/(1 - z)$ is consistent with the same original clock. These statements concern unweighted counting.

3 One complex disk for countably many branches

Fix once and for all

$$\Omega = \{z : |z - 1| < 3/4\}, \quad e_k(z) = \left(\frac{z-1}{3/4}\right)^k \quad (k \geq 0), \quad (8)$$

where the e_k are the normalized monomial basis of $H^2(\Omega)$. The notation $D(c, r)$ denotes the open disk of center c and radius r .

Lemma 1 (Strict common range and reciprocal growth). *Every h_n is holomorphic on a neighborhood of $\overline{\Omega}$ and*

$$h_n(\Omega) \subset D(1, 1/2) \Subset \Omega, \quad \sup_{\Omega} |h'_n| \leq 1/2. \quad (9)$$

For $w_m = g^m(z)$, uniformly on Ω ,

$$|w_m| \leq \frac{1}{4/7 + 2m/25} \leq \frac{25}{2(m+1)}. \quad (10)$$

Proof. If $\operatorname{Re} z > 0$, then $\operatorname{Re} \sqrt{1+8z} > 1$, so $w = g(z)$ lies in the right half-plane. The identity $z = w(1+2w)$ yields

$$\frac{1}{w} = \frac{1}{z} + \frac{2}{1+2w}. \quad (11)$$

The added term has positive real part. The disk $D(1, 1)$ is exactly the set of nonzero z with $\operatorname{Re}(1/z) > 1/2$; hence g preserves it. Starting from Ω , every w_m stays in that disk. For $n \geq 2$ this gives $|h_n(z) - 1| < 1/2$, while for $n = 1$ the bound is $3/8$. The closed initial disk stays inside the right half-plane, where all iterated branches are holomorphic. The estimate $|g'| = |1+8z|^{-1/2} \leq 1$ there gives the derivative bound in (9); convexity also gives a Lipschitz constant $1/2$ on the closed disk.

For $w \in D(1, 1)$, $|w| \leq 2$ and $\operatorname{Re} w > 0$. Thus $\operatorname{Re}(2/(1+2w)) \geq 2/25$. The initial disk has $\operatorname{Re}(1/z) \geq 4/7$. Iteration of (11) proves (10). $\square$

Theorem 3 (All-branch derivative decay). *An explicit uniform constant is*

$$C_* = 1250 \exp(625\pi^2/6), \quad \sup_{\Omega} |h'_n| \leq \frac{C_*}{n^2} \quad (n \geq 1). \quad (12)$$

Proof. Differentiate $w_{k-1} = w_k(1+2w_k)$ and telescope to obtain

$$\begin{aligned} (g^m)'(z) &= \left(\frac{w_m}{z}\right)^2 \prod_{k=1}^m \frac{(1+2w_k)^2}{1+4w_k} \\ &= \left(\frac{w_m}{z}\right)^2 \prod_{k=1}^m \left(1 + \frac{4w_k^2}{1+4w_k}\right). \end{aligned} \quad (13)$$

Since $|1+4w_k| \geq 1$, the absolute product is at most $\exp(4 \sum_{k \geq 1} |w_k|^2)$. Equation (10) bounds its exponent by $625\pi^2/6$. Also $|z| \geq 1/4$, and with $m = n-1$ the leading square is at most $2500/n^2$. The factor $1/2$ in h_n gives (12). Its intentionally large constant proves convergence; it is not advertised as a useful numerical cutoff certificate. $\square$

4 A trace-class operator with two explicit clocks

Let u count returns to Y , and let ζ count original applications of T . Define

$$A_n f = h'_n f \circ h_n, \quad \mathcal{L}_\zeta = \sum_{n \geq 1} \zeta^n A_n, \quad \Delta(u, \zeta) = \det(I - u \mathcal{L}_\zeta). \quad (14)$$

Theorem 4 (Nuclear inducing on the closed original-clock disk). *The branch series converges absolutely in trace norm on $H^2(\Omega)$ for every $|\zeta| \leq 1$. It is trace-norm continuous on that closed disk and holomorphic inside it. For fixed such ζ , Δ is entire in u ; it is jointly holomorphic for $u \in \mathbb{C}$, $|\zeta| < 1$. The displayed branch series fails to converge for $|\zeta| > 1$ even on the constant function. This last assertion does not exclude separately proved analytic continuation.*

Proof. Write $f = \sum c_k e_k$. The coefficient functional has norm one and strict containment in (9) gives

$$\|h'_n e_k \circ h_n\|_{H^2} \leq \sup_{\Omega} |h'_n| (2/3)^k. \quad (15)$$

Thus the explicit rank-one expansion has summable norms and $\|A_n\|_1 \leq 3 \sup |h'_n| \leq 3C_*/n^2$. Hilbert-space nuclearity here is trace class. Summing proves the norm convergence and boundary continuity; locally uniform trace-norm convergence proves interior holomorphy. The ordinary trace-class Fredholm determinant gives the asserted parameter regularity and entire dependence on u .

For any real $x \in [1/2, 1]$, the real reciprocal increment is at most two, so $g^{n-1}(x) \geq 1/(2n)$. Every real factor in (13) is at least one, and $x \leq 1$, yielding $h'_n(x) \geq 1/(8n^2)$. If $|\zeta| > 1$, the terms $\zeta^n h'_n(x)$ do not tend to zero. The branch series therefore fails the necessary term test already for $f = 1$. $\square$

Theorem 5 (Primitive determinant with the original and induced times). *For a primitive induced cycle p , let r_p be its number of returns, N_p its total original time, and $\Lambda_p > 1$ its forward multiplier. For $|\zeta| \leq 1$, set*

$$K(\zeta) = \sum_{n \geq 1} |\zeta|^n \sup_{\Omega} |h'_n| < \infty. \quad (16)$$

For $|u|K(\zeta) < 1$, the absolutely regrouped primitive product is

$$\Delta(u, \zeta) = \prod_{p \text{ primitive induced}} \prod_{k \geq 1} \left(1 - \frac{u^{r_p} \zeta^{N_p}}{\Lambda_p^k} \right). \quad (17)$$

At $u = 1$ this identity holds initially for sufficiently small $|\zeta|$. Entire continuation in u of the determinant does not imply unrestricted convergence of its primitive product.

Proof. Every r -branch composition χ maps the closed disk into the common interior disk and has Lipschitz constant at most 2^{-r} . It has exactly one fixed point x_χ , and invariance of $[1/2, 1]$ makes that point real. Write $C_\chi f = f \circ \chi$ for composition by this branch. Summing diagonal Hardy coefficients as a geometric series and applying the residue theorem gives

$$\text{tr}(\chi' C_\chi) = \frac{1}{2\pi i} \int_{|z-1|=3/4} \frac{\chi'(z)}{z - \chi(z)} dz = \frac{\chi'(x_\chi)}{1 - \chi'(x_\chi)}. \quad (18)$$

Strict containment justifies summation under the contour. Contraction gives the unique zero of $z - \chi(z)$ and makes it simple. Summing the absolute trace contributions over all compositions gives the bound $2K(\zeta)^r$.

First use the trace-log determinant identity on the smaller domain $|u||\mathcal{L}_\zeta| < 1$. The absolute coefficient bound extends its scalar logarithmic series to $|u|K(\zeta) < 1$. The identity theorem then identifies its exponential with the entire determinant throughout that disk. This argument does not identify K with the Hardy operator norm.

For an a -fold repeat of p , the inverse multiplier is Λ_p^{-a} , so its trace weight is

$$\frac{\Lambda_p^{-a}}{1 - \Lambda_p^{-a}} = \sum_{k \geq 1} \Lambda_p^{-ak}, \quad (19)$$

and its two clocks are (ar_p, aN_p) . Grouping r_p based phases and dividing by ar_p in the logarithm leaves $1/a$. Absolute convergence now permits the primitive and repetition regrouping, giving (17). Finally $K(\zeta) \rightarrow 0$ as $\zeta \rightarrow 0$, which gives the stated $u = 1$ specialization. $\square$

The neutral fixed point has multiplier one and would make the flat-trace denominator vanish. It cannot be inserted as another ordinary trace-class branch contribution. Equation (7) and the derivative-weighted determinant (17) are distinct objects even though both retain the original iteration clock.

Literature ownership and claim boundary

The intermittent-map lineage is represented by Liverani, Saussol and Vaienti [1]. The alpha-one endpoint is treated by the explicit calculations here; no finite stationary-measure theorem is imported from a different parameter range. Rugh's work [2] establishes the classical connection between intermittency and regularized Fredholm determinants on carefully specified function spaces. The present package neither claims to discover inducing nor replaces those function-space qualifications. Its increment is the connected audit of one literal endpoint convention, complete primitive coding, sharp reference return tail, and one all-branch complex domain with two determinant clocks. No worldwide novelty or publication-acceptance claim is made.

References

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Round one: common complex domain and induced Fredholm product